ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

ELECTRICAL MACHINES

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 2nd year final examination: magnetic circuits, single-phase transformers (equivalent circuit, efficiency and regulation), three-phase circuits and the per-unit system, DC machines and three-phase induction motors.

Written by Academic Central for practice, following electrical machines modules at South African universities (e.g. University of Pretoria ELX 311 Electrical Machines and EIR 211 Electrical Engineering: magnetic circuits, transformers, per-unit system, three-phase circuits, DC and induction motors). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Show ALL your working, including formulae and substitutions. Give answers correct to TWO decimal places, with units.
3. Use $\mu_0 = 4\pi \times 10^{-7}$ H/m. Three-phase quantities are line values unless stated otherwise.
4. Non-programmable scientific calculators may be used.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: MAGNETIC CIRCUITS

1.1 Define magnetomotive force (MMF), magnetic flux, flux density and reluctance, and give the unit of each. (4)

1.2 An iron ring has a mean length of 0.4 m, a cross-sectional area of 5 cm² and a relative permeability of 1 000. It carries a coil of 200 turns.

1.2.1 Calculate the reluctance of the ring and the current needed to produce a flux of 0.5 mWb. (4)

1.2.2 Calculate the flux density in the ring. (1)

1.3 A 1 mm air gap is cut in the ring. Ignore fringing and the small change in iron length. Calculate the current now needed for the same flux. (4)

1.4 Explain hysteresis loss and eddy-current loss in an iron core, and how each is reduced in practice. (4)

1.5 The flux linking a 500-turn coil changes uniformly by 2 mWb in 10 ms. Calculate the induced EMF. (3)

[20]

QUESTION 2: SINGLE-PHASE TRANSFORMERS

2.1 Explain how a transformer works. (3)

2.2 An ideal 230 V / 12 V, 50 VA transformer has 1 200 primary turns.

2.2.1 Calculate the turns ratio and the number of secondary turns. (3)

2.2.2 Calculate the full-load primary and secondary currents. (2)

2.3 Use the EMF equation to calculate the peak flux in the core of the transformer in 2.2 (f = 50 Hz). (3)

2.4 Draw the approximate equivalent circuit of a practical transformer referred to the primary, and say what each element represents. (4)

2.5 A 10 kVA, 2 000 V / 200 V transformer has a core loss of 200 W (open-circuit test) and a full-load copper loss of 300 W (short-circuit test).

2.5.1 Calculate the efficiency at full load and a power factor of 0.8. (3)

2.5.2 Calculate the efficiency at half load and a power factor of 0.8. (3)

2.5.3 At what fraction of full load is the efficiency a maximum? (2)

2.6 The secondary voltage of the transformer in 2.5 is 200 V on no load and 192 V on full load. Calculate the voltage regulation. (2)

[25]

QUESTION 3: THREE-PHASE CIRCUITS AND THE PER-UNIT SYSTEM

3.1 A balanced star-connected supply has a line voltage of 400 V. State the phase voltage and the relationship between line and phase currents. Then state both relationships for a delta connection. (4)

3.2 A balanced three-phase load takes 15 kW at 400 V and a power factor of 0.85 lagging. Calculate the line current, the apparent power and the reactive power. (5)

3.3 A 400 V, three-phase system uses a base of 10 kVA and 400 V. Calculate the base current and the base impedance, and express an impedance of 0.8 Ω in per unit. (4)

3.4 Give TWO advantages of using the per-unit system. (2)

[15]

QUESTION 4: DC MACHINES

4.1 Explain the purpose of the commutator and brushes in a DC motor. (3)

4.2 A separately excited DC motor runs from 220 V. Its armature resistance is 0.5 Ω, and at 1 000 r/min it draws an armature current of 20 A.

4.2.1 Calculate the back EMF. (2)

4.2.2 Calculate the power developed in the armature and the torque developed. (3)

4.2.3 The load increases and the armature current rises to 30 A with the same field. Calculate the new speed. (3)

4.3 Calculate the starting current of the motor in 4.2 if it is switched straight on, explain why this is a problem, and calculate the starting resistance needed to limit the starting current to 40 A. (4)

4.4 Describe TWO methods of controlling the speed of a DC motor, and say which one a robot motor driver normally uses. (3)

4.5 Compare the torque-speed behaviour of a DC series motor and a DC shunt motor, and give one application of each. (2)

[20]

QUESTION 5: THREE-PHASE INDUCTION MOTORS

5.1 Explain how a rotating magnetic field makes the rotor of a squirrel-cage induction motor turn. (3)

5.2 A 4-pole, 50 Hz induction motor runs at 1 440 r/min. Calculate the synchronous speed, the slip and the frequency of the rotor currents. (4)

5.3 The motor in 5.2 delivers 7.5 kW and draws 15 A from a 400 V supply at a power factor of 0.85. Calculate the output torque, the input power and the efficiency. (5)

5.4 The air-gap power of the motor is 8 kW. Calculate the rotor copper loss and the mechanical power developed. (3)

5.5 Compare direct-on-line starting and star-delta starting of an induction motor. (3)

5.6 A variable-speed drive keeps the V/f ratio constant. Calculate the supply frequency and voltage needed for a synchronous speed of 1 200 r/min. (2)

[20]

TOTAL: 100

END OF QUESTION PAPER